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Conditions (skin, interface, border, surface)

Boundary Conditions

$$\mathbf{n} \cdot [\mathcal{D}] = \sigma$$

$$\mathbf{n} \cdot [\mathcal{B}] = 0$$

$$\mathbf{n} \times [\mathcal{E}] = 0$$

$$\mathbf{n} \times [\mathcal{H}] = J$$

Interior Conditions

e.g. for Superconductivity

$$\mathbf{n} \cdot \mathcal{H} = 0$$

$$\mathbf{n} \times \mathcal{D} = 0$$

Abbreviations

$\mathbf{n}$ = normal vector (orthogonality orientation)

$\mathcal{E}$ = elec. Field (“traction force?”)

$\mathcal{B}$ = magn. Field (“compressive force?”)

$\mathcal{D}$ = elec. Displacement Field (“exitioan”)

$\mathcal{H}$ = magn. Exitation Field (“incentivation”)

σ = electric charge at surface

J = current density at surface